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8 February 1977
SUPERSEDING
MIL-C-36843
6 June 1972

MILITARY SPECIFICATION

CUP, DENTAL AMALGAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a metal cup suitable for holding amalgam.

1.2 Classification. The cups covered by this specification shall be of the following styles, as specified (see 6.1):

Style 1 - Cup, with rubber washer.

Style 2 - Cup, without washer.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 S. 20th Street, Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

23245	-CUP, DENTAL AMALGAM.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 2240 - Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General requirements. The cups shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

3.2 Material.

3.2.1 Metal. Cups shall be fabricated of class 3, type 303 stainless steel, class 4, type 416 stainless steel, or 416 +x stainless steel conforming to GG-I-526, except 416 +x shall have a maximum manganese content of 2.50 percent.

3.2.2 Rubber. The flat washer shall be fabricated of black rubber suitable for attachment to base of style 1 cup (see 3.3.1).

3.3 Style, design and dimensions. The metal cup shall be suitable for holding triturated amalgam to facilitate amalgam carrier loading. The design and dimensions of the cups shall conform to Defense Personnel Support Center (DPSC) Drawing No. 23245 for style 1 or 2, as specified.

3.3.1 Style 1 cup. The diameter of the base shall be larger than the diameter of the receptacle area to prevent tipping. A flat rubber washer shall fit in a groove on the circumference of the base of the metal cup to prevent skidding. The external top circumference area of the cup shall be knurled to facilitate handling.

3.3.2 Style 2 cup. The cup shall be cylindrical in shape. The external curved surface shall contain two parallel grooves, equidistant from each end to facilitate handling.

3.4 Hardness. Durometer hardness of the flat rubber washer attached to style 1 cup shall be 65 $\pm$ 5 points, when tested as specified in 4.3.1.

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3.5 Weight. The weight of the style 1 cup with attached flat washer shall be 110.0 grams minimum, and the weight of the style 2 cup shall be 150.0 grams minimum, when tested as specified in 4.3.2.

3.6 Finish. The internal cavity and the outside surface of the cup shall have a mirror or bright finish, except for the knurled area of style 1 cup. The flat bottom surface of the cup shall be smooth. The cup shall be furnished clean and ready for use.

3.7 Identification marking. The metal cup shall be permanently and legibly marked with the name or registered trademark of the manufacturer. Marking shall appear on the flat bottom surface of the cup.

3.8 Workmanship. Workmanship shall be first class throughout. Cups shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metal, rubber and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the date of testing, the test method, individual test results, the specification requirements, the lot number, the lot size and the sample size.

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4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Cups shall be examined for defects, including, but not limited to, those listed in table I.

TABLE I. Classification of defects.

Categories	Defects
Critical:	None defined
Major:	
101	Style and design not as specified.
102	Finish of internal surface of cup not as specified.
103	Dimensions of flat washer and the groove on the circumference of the base of the style 1 metal cup not as specified.
Minor:	
201	Finish (other than 102) not as specified.
202	Dimensions (other than 103) not as specified.
203	Cup not permanently and legibly marked with name or trademark of manufacturer.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table II. The unit of product for examination shall be one cup.

TABLE II. Examination criteria.

Type Defect	Inspection Level	AQL (Percent Defective)
Major	S-2	1.5
Minor	S-2	4.0

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4.2.2 Testing. Cups shall be tested as specified in table III and 4.3. Inspection levels and acceptable quality levels (AQL's) shall be as listed in table III. The unit of product for testing shall be one cup.

TABLE III. Test criteria.

Characteristic	Component	Requirement	Test procedure	Inspection Level	AQL (Percent defective)
Composition Analysis Style 1 and 2	Cup	3.2.1 and GG-I-526	GG-I-526	**	*
Hardness Style 1	Flat washer	3.4	4.3.1	S-1	1.5
Weight Style 1	Cup plus flat washer	3.5	4.3.2	S-1	1.5
Style 2	Cup	3.5	4.3.2	S-1	1.5
Corrosion Resistance Style 1 and 2	Cup	GG-I-526	GG-I-526***	S-2	1.0

* Acceptance number shall be zero for all sample sizes.

** A sufficient amount from each lot of metal used in the construction of the cups shall be selected for chemical analysis.

*** Boil test for class 3 stainless steel; boil and copper sulfate tests for class 4 stainless steel.

4.2.3 Packaging inspection. The inspection of the unit package, intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Hardness. Hardness of flat washer shall be determined with a Shore A-2 durometer in accordance with ASTM Method D 2240.

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4.3.2 Weight. The weight of the style 1 cup with the attached flat washer, and style 2 cup, shall be determined using a balance suitable for the intended purpose.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each cup shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be in accordance with the appendix of the applicable box specification.

5.1.1.2 Intermediate packaging. Twelve (12) cups shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be in accordance with the appendix of the box specification.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. One hundred forty-four (144) cups shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with the appendix of PPP-B-636.

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5.2.2 Level B. One hundred forty-four (144) cups shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit, intermediate and exterior container. Each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Style required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog: 6520-00-138-9748, CUP, DENTAL AMALGAM, Corrosion-Resisting Steel.

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6520-1430

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-C-36843A CUP, DENTAL AMALGAM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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